



# Shiftbase QA Assessment

End-to-end automation of the signup & login flows

**Both flows automated to the dashboard — with a handful of observations the tests surfaced on the demo environment along the way.**

Playwright + TypeScript, on the element-interactions Steps API. Runs green from a cold clone with no configuration.

# The brief, and what shipped

Automate two flows on the Shiftbase demo environment: signup + onboarding up to the dashboard, and login. This is what came back.



## Both flows, end to end

Signup walks the 8-step questionnaire and setup wizard to the dashboard; login handles the MFA interstitial. Interactions read as intent through the Steps API — no raw selectors in any test.



## Some observations along the way

Testing a real product surfaces things. A few came up while building the happy paths, and a focused exploratory pass found more — all reproduced, none false positives. Shared constructively, for the team to weigh.



## Evidence, not claims

Each observation carries a reproduction and a screen recording or screenshot. Anything that couldn't be pinned by a portable test is written up with the reason, not forced into one.



## Runnable by anyone

`npm ci && npm test` on a clean clone with no `.env`. Verified by cloning fresh; the suite provisions its own account through the real signup flow.

# Coverage at a glance

29 tests across three Playwright projects: desktop, mobile, and an account-provisioning setup project.

23

passing tests

6

failing by design

6

pages mapped

56

elements

11

observations



## The 6 failing tests fail on purpose

Each is tagged `@known-defect` and describes the behaviour the product is expected to have — so it fails today and would turn green once addressed, becoming a regression guard. `npm run test:no-defects` is 23 / 23 green.



## Not weakened to pass

Rewriting a test to agree with unexpected behaviour would make the suite unable to catch it. The failing tests were left describing the expected behaviour instead.

# What each test protects

Chosen, not sprayed — every scenario covers a distinct way a flow can break.

| FLOW    | COVERAGE                                                                                                                                                                                                                                    |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signup  | Registration → questionnaire → wizard → dashboard · form handoff · mobile viewport · empty-submit & per-field validation (email, phone, weak password, terms)                                                                               |
| Login   | Valid credentials · the MFA-promote interstitial · wrong password · unknown email · <b>identical rejection message (no account enumeration)</b> · protected-route redirect · “Back to login” ends the session · concurrent & stale sessions |
| Session | Survives a reload (asserts the dashboard re-renders, not just the URL) · primary navigation present                                                                                                                                         |

Deliberately out of scope, and noted in the README: everything past the dashboard, password reset, cross-browser, visual regression, load.

# What the tests surfaced

Eleven observations on the demo environment, each reproduced and independently re-checked. Shared as the kind of thing thorough automation is built to catch.

Highest priority · consent

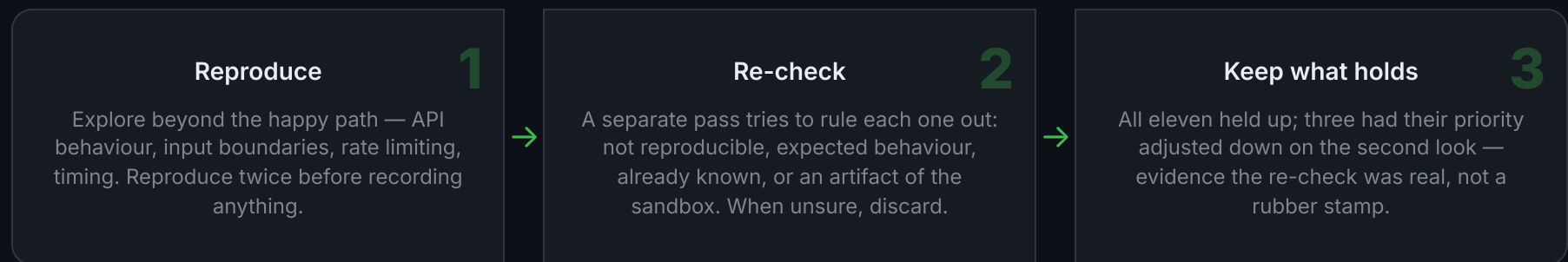
## Terms & Data-Processing-Agreement acceptance is enforced only in the browser

On the demo environment, `POST /api/signup` accepts a request with `accept_conditions: false` and creates a usable account — the consent checkbox the UI requires has no server-side counterpart. Worth a look because the same pattern in production would record consent that was never given. Reported rather than automated, so a regression run never repeats it against the shared environment.

| SOURCE           | OBSERVATIONS                                                                                                                                                                                                              | COVERAGE               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| While building   | Blank page on an unrecognised modal route · duplicate email accepted, then its password rejected at login · no length cap on the name field                                                                               | 2 of 3 as tests        |
| Exploratory pass | Consent bypass · no login rate-limit · timing enumeration · gateway error shown as bad credentials · swallowed reset message · fractional employee count · missing Back button · unstructured 500 on a malformed API call | 2 as tests, 6 reported |

# How each observation was verified

Nothing rests on a single look — each was reproduced, then re-checked by an independent pass whose job was to disprove it.



- ✓ **Selectors that survive copy changes**  
The questionnaire's radio inputs aren't directly clickable (`sr-only`); options are targeted by their stable `formcontrolname + value`, not the visible label — which can differ from the value.
- ✓ **Unexpected behaviour reported, tests left honest**  
Where the app behaved unexpectedly, the test was left describing the expected behaviour and the observation was written up — never adjusted to match.

# Engineering choices worth a look

The decisions a reviewer running this cold would notice.



## No committed credentials

A setup project provisions an account via the real signup flow when no `.env` is present, so a cold clone runs green. Secrets live in `.env` (gitignored); an independent sweep confirmed none shipped.



## Flake, root-caused not retried

The Angular questionnaire can drop clicks mid-render; every input is re-applied until it lands (verified 12× with retries off). The residual is the environment's own gateway variability, left to configured retries.



## Watchable headed runs

`npm run test:headed` drops to one worker and paces each action, so a reviewer can actually follow the flows rather than watch windows race.



## TLS reality documented

The demo serves a private-CA cert that rolls every 24h; the suite sets `ignoreHTTPSErrors` and says so plainly — on a production target that setting would hide a real issue.

# Ready to run, honest about its edges

The README carries the full reasoning, the observations, and a candid “what I’d do next.”



## What I’d do next

Flag the server-side consent check first — a small change with outsized value. Move account provisioning to the API to cut runtime. Add teardown for the accounts each run leaves behind.

@playwright/test

1.62

@civitas-cerebrum/element-interactions

0.3.9

TypeScript

strict